SAFETY IN DESIGN

1. THIS DOCUMENT OUTLINES POTENTIAL HEALTH AND SAFETY RISKS RELATED TO THE STRUCTURAL ASPECTS OF THIS PROJECT DURING CONSTRUCTION, OPERATION, MAINTENANCE AND DECOMMISSIONING.
2. IT IS ASSUMED THAT CONTRACTORS AND SUBCONTRACTORS ARE QUALIFIED AND TRAINED TO RECOGNISE AND MANAGE TYPICAL SITE RISKS.
3. IT IS ASSUMED END USERS ARE TRAINED IN SAFE OPERATION AND MAINTENANCE OF THE FACILITY, AND ANY SPECIFIC REQUIREMENTS HAVE BEEN PROVIDED FOR CONSIDERATION IN THE DESIGN.
4. THIS INFORMATION IS TO BE READ ALONGSIDE INPUTS FROM OTHER DESIGNERS.
5. ANY HAZARDS CONSIDERED UNUSUAL OR SITE-SPECIFIC ARE LISTED IN THE ACCOMPANYING TABLE. CONTRACTORS AND SUBCONTRACTORS ARE EXPECTED TO REVIEW ALL WORKS IN THEIR SCOPE AND ADVICE OF ANY OTHER SAFETY ISSUES NOT IDENTIFIED HERE.
6. THESE NOTES DO NOT REMOVE OBLIGATIONS UNDER THE WORK HEALTH AND SAFETY ACT 2011 (QLD), INCLUDING CONSULTATION WITH THE DESIGNER, PREPARATION OF SAFE WORK METHODS STATEMENTS, AND FULFILLING DUTIES OF CARE. SAFETY DURING CONSTRUCTION REMAINS THE CONTRACTOR'S RESPONSIBILITY.
7. THE CONTRACTOR SHALL COMPLETE AND WILL SOLEY BE RESPONSIBLE FOR THE IMPLEMENTATION OF ANY NECESSARY SAFETY PLANS TO COMPLETE THE WORKS.
8. THE CONTRACTOR MUST COMPLY WITH ANY GUIDELINE ACTS OR CODE OF PRACTICE AND OTHER RELEVANT DOCUMENTS REGARDING SAFE WORK PRACTICES.
9. THE CONTRACTOR SHALL HAVE A NOMINATED WORKPLACE HEALTH AND SAFETY (WH&S) OFFICER FOR THE DURATION OF THE CONTRACT. THE WH&S OFFICER WILL BE RESPONSIBLE FOR ALL WH&S ISSUES ON SITE.

SAFETY IN DESIGN GENERAL NOTES

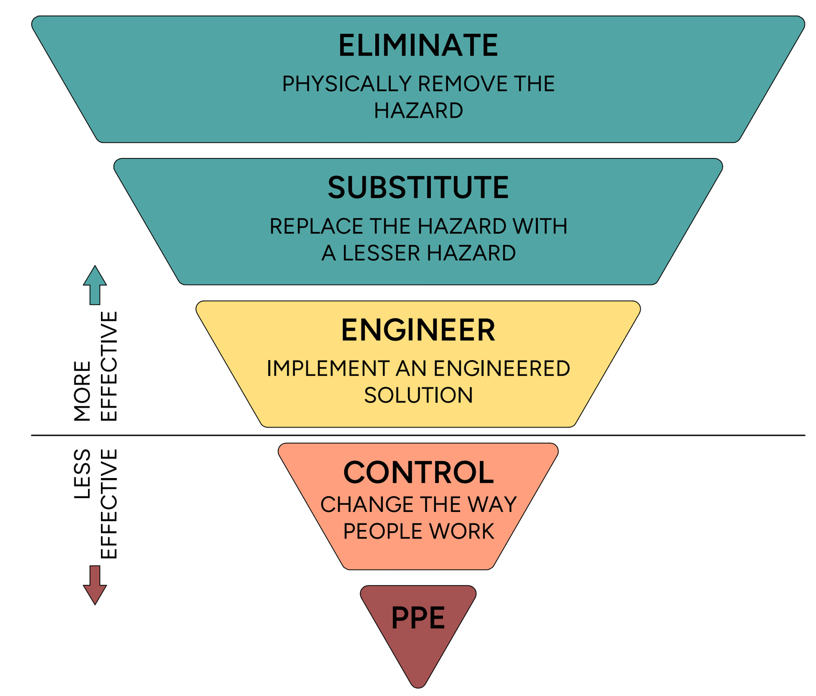
CONSULT	INSPECT	IDENTIFY	MITIGATE	REVIEW
ESTABLISH DESIGN CONTEXT AND CONSULTATION METHOD WITH THE CLIENT	CONDUCT A PRELIMINARY HAZARD ANALYSIS AND CONSULTATION	IDENTIFY HAZARDS INFLUENCED BY THE DESIGN AND WITHIN CONTROL OF THE DESIGNER	DETERMINE HOW HAZARDS WILL BE ELIMINATED OR MINIMIZED	REVIEW DESIGN TO VERIFY WHETHER RISK ELIMINATION OR REDUCTION HAS BEEN ACHIEVED

CONSEQUENCE \ LIKELIHOOD	1 - RARE MAY OCCUR IN EXCEPTIONAL CIRCUMSTANCES	2 - UNLIKELY COULD OCCUR AT SOME TIME	3 - POSSIBLE MIGHT OCCUR AT SOME TIME	4 - LIKELY WILL OCCUR IN MOST CIRCUMSTANCES	5 - ALMOST CERTAIN EXPECTED IN MOST CIRCUMSTANCES
5 - CATASTROPHIC DEATH, PERMANENT INJURY OR LASTING DAMAGE	5	10	15	20	25
4 - MAJOR EXTENSIVE INJURIES, DAMAGE OR LOSS OF PRODUCTION	4	8	12	16	20
3 - MODERATE REQUIRING MEDICAL TREATMENT OR OUTSIDE ASSISTANCE	3	6	9	12	15
2 - MINOR FIRST AID ONLY OR DAMAGE CONTAINED TO SITE	2	4	6	8	10
1 - NEGLIGIBLE NO INJURIES OR DAMAGE	1	2	3	4	5

RISK RATING MATRIX

1-4	LOW *NO FURTHER ACTION REQUIRED UNLESS BENEFICIAL ACTION CAN BE EASILY UNDERTAKEN*
5-9	MODERATE *CONTROL MEASURES MUST BE PUT IN PLACE TO REDUCE THIS RISK UNLESS IT WOULD INVOLVE EXCESSIVE COST FOR LITTLE BENEFIT*
10-19	HIGH *ALL REASONABLY PRACTICABLE CONTROL MEASURES MUST BE PUT IN PLACE BEFORE THIS TASK MAY PROCEED*
20+	VERY HIGH *THIS ACTIVITY MUST NOT BE UNDERTAKEN, OR SHOULD BE STOPPED UNTIL ADDITIONAL MEASURES HAVE BEEN PUT IN PLACE TO REDUCE THE RISK*

CONSEQUENCE TABLE



CONTROL HEIRARCHY

ITEM	RISK	PRE-CONTROL ASSESSMENT			CONTROL	RESPONSIBILITY	POST-CONTROL ASSESSMENT			RESIDUAL RISK OWNER
		CONSEQUENCE	LIKELIHOOD	RISK RATING			CONSEQUENCE	LIKELIHOOD	RISK RATING	
SITEWORKS										
1.1	OVERHEAD OR UNDERGROUND SERVICES IMPACT ON STRUCTURE	3	3	9	SURVEY OF EXISTING SERVICES, DESIGN AROUND THEM TO AVOID CLASHES.	BUILDER	2	3	6	BUILDER
1.2	DAMAGE TO IN GROUND EXISTING SERVICES	3	4	12	THOROUGH SITE INVESTIGATION PRIOR TO COMMENCEMENT OF WORK. NOTIFY THE DESIGN TEAM OF ANY DISCREPANCIES OR NEW SERVICES FOUND. REVIEW ALL AVAILABLE DOCUMENTATION ON EXISTING SERVICES AND INCORPORATE MEASURES TO AVOID OR ELIMINATE POTENTIAL FOR DAMAGE. ADD LOCATION OF SERVICES TO DOCUMENTATION	BUILDER	2	4	8	BUILDER
1.3	PROPOSED STRUCTURE IMPACTS ON ADJACENT PROPERTIES, EG. EXCAVATION DEPTHS NEAR BOUNDARIES OR EXISTING STRUCTURES (UNDERMINING)	3	2	6	APPROPRIATE DESIGN SOLUTIONS TO AVOID/MINIMISE IMPACT ON STRUCTURES OR PROPERTIES. NEW SERVICES TO BE LOCATED SUCH THAT UNDERMINING OF ADJACENT STRUCTURES DOES NOT OCCUR. WHERE THIS IS NOT POSSIBLE, INCORPORATE MEASURES SUCH AS UNDERPINNING TO ENSURE SHORT AND LONG TERM STABILITY	STRUCTURAL ENGINEER SERVICES ENGINEER BUILDER	2	2	4	STRUCTURAL ENGINEER SERVICES ENGINEER BUILDER
1.4	INAPPROPRIATE STRUCTURAL DESIGN FOR FOUNDING MATERIAL. RISK OF SETTLEMENT, FAILURE	3	3	9	CARRY OUT THOROUGH GEOTECHNICAL INVESTIGATION AND BASE DESIGN OF STRUCTURE AND FOUNDATIONS ON THE GEOTECHNICAL ADVIC. CONTRACTOR TO ENGAGE SUITABLE GEOTECHNICAL CONSULTANTS TO CONFIRM IT AS PER STRUCTURAL ENGINEERS DESIGN CONDITIONS. FOLLOW GEOTECHNICAL REPORT FOR SLOPE STABILITY REQUIREMENTS. SLOPE STABILITY ASSESSMENT TO BE ENGAGED AND ACTIVE ONGOING GEOTECHNICAL INSPECTIONS TO BE CARRIED OUT.	STRUCTURAL ENGINEER SERVICES ENGINEER BUILDER	2	3	6	STRUCTURAL ENGINEER SERVICES ENGINEER BUILDER
1.5	FALL WHILE WORKING ADJACENT DEEP EXCAVATIONS, OR AT HEIGHTS	3	5	15	ADEQUATE SAFE WORK PRACTICES INCLUDING FALL PROTECTION/ARREST SHOULD BE PROVIDE BUT IS NOT PROVIDED BY STRUCTURAL ENGINEER AND IS THE RESPONSIBILITY OF THE BUILDER. FOR DESIGN OF FIXINGS AND SIMILAR FOR SUCH SYSTEMS, SEPARATELY ENGAGE A STRUCTURAL ENGINEER AS REQUIRED.	BUILDER	1	5	5	BUILDER
1.6	COLLAPSE OR INSTABILITY OF TEMPORARY WORKS (E.G. FORMWORK, PROPPING, STRIPPING, MASONRY WALLS, TILT-UP WALLS)	3	4	12	THE STRUCTURAL ENGINEERS DESIGN IS FOR IN SERVICE LOADS ONLY. BUILDER TO ENGAGE SUITABLY QUALIFIED TEMPORARY WORKS ENGINEER FOR DESIGN OF TEMPORARY PROPPING WORKS DURING CONSTRUCTION. ENGINEER TO PROVIDE A CERTIFIED DESIGN FOR TEMPORARY WORK.	BUILDER TEMPORARY WORKS ENGINEER	1	4	4	BUILDER TEMPORARY WORKS ENGINEER
1.7	COLLAPSE OF RETAINING STRUCTURES	3	4	12	RETAINING WALLS TO BE PROPPED DURING CONSTRUCTION IF REQUIRED BY ENGINEERS DOCUMENTATION. TEMPORARY PROPPING TO BE DESIGNED AND CERTIFIED BY TEMPORARY WORKS ENGINEER. VEHICLES/HEAVY MATERIALS TO MAINTAIN SAFE DISTANCE FROM BEHIND TOP OF WALL WITH FENCED EXCLUSION ZONE AS REQUIRED TO ENSURE WALL IS NOT OVERLOADED.	GEOTECHNICAL ENGINEER TEMPORARY WORKS ENGINEER BUILDER	1	4	4	GEOTECHNICAL ENGINEER TEMPORARY WORKS ENGINEER BUILDER
1.8	OVERLOAD OF STRUCTURE DURING CONSTRUCTION	3	5	15	STRUCTURAL DOCUMENTATION TO INCLUDE DESIGN LOADING LIMITATIONS. ALL CONSTRUCTION LOADING TO COMPLY WITH DESIGN LOADING LIMITATIONS. BUILDER TO SEEK APPROVAL FROM THE STRUCTURAL ENGINEER PRIOR TO ANY ADDITIONAL LOADS APPLIED TO THE STRUCTURE.	STRUCTURAL ENGINEER BUILDER	2	5	10	STRUCTURAL ENGINEER BUILDER
1.9	STABILITY OF BATTERS TO STRUCTURAL ELEMENTS	3	5	15	GEOTECHNICAL AND CIVIL REPORTS, DRAWINGS, AND ADVICE SHOULD BE FOLLOWED FOR THE SAFE AND RECOMMENDED BATTER ANGLES AND LOCATIONS. CONTRACTOR TO INSTALL TEMPORARY PHYSICAL BARRIERS (SEPARATION ZONES, MULCH/DIRT BUNDS, FLAGGING/FENCING) AS REQUIRED UNDER THEIR SAFETY MANAGEMENT PLAN WHILE WORKING AT THE TOP OF ANY STEEP BATTERS. REMOVAL OF THE ABOVE CONTROLS TO BE CAREFULLY TIMED AND PLANNED TO AVOID CONSTRUCTION PLANT OPERATING ABOVE BATTER ONCE CONTROLS ARE REMOVED.	CIVIL ENGINEER GEOTECHNICAL ENGINEER BUILDER	2	5	10	CIVIL ENGINEER GEOTECHNICAL ENGINEER BUILDER
REINFORCED CONCRETE WORKS										
2.1	COLLAPSE OR DAMAGE TO STRUCTURE DUE TO OVERLOADING OF CONSTRUCTION MATERIAL OR PLANT	4	4	16	DESIGN DRAWINGS TO CLEARLY INCLUDE ALLOWABLE LOADS ON DOCUMENTATION. CONTRACTOR TO ADHERE TO ALLOWABLE LIMITS.	STRUCTURAL ENGINEER BUILDER	2	4	8	STRUCTURAL ENGINEER BUILDER
2.2	COLLAPSE/DAMAGE TO STRUCTURE OR EXISTING SERVICES DUE TO UNAUTHORISED CUTTING, CORING OR DRILLING TO ACCOMMODATE NEW SERVICES	3	5	15	ALL POST-INSTALLED PENETRATIONS/CORING TO EXISTING STRUCTURAL ELEMENTS THAT ARE NOT EXPLICITLY SHOWN ON STRUCTURAL DOCUMENTATION ARE TO BE APPROVED BY A SUITABLY QUALIFIED STRUCTURAL ENGINEER	BUILDER CLIENT SERVICES ENGINEER	1	5	5	BUILDER CLIENT SERVICES ENGINEER
2.3	INADEQUATE BACKPROPPING OR EARLY STRIPPING OF FORMWORK	3	4	12	FOLLOW REQUIREMENTS OUTLINED IN STRUCTURAL SPECIFICATION AND AS3600:2018. PROPPING TO BE DESIGNED AND CERTIFIED BY SUITABLY QUALIFIED TEMPORARY WORKS ENGINEER	BUILDER TEMPORARY WORKS ENGINEER	2	4	8	BUILDER TEMPORARY WORKS ENGINEER
2.4	FAILURE OF POST-FIXED ANCHORS (CHEMICAL/MECHANICAL)	4	4	16	AVOID PERMANENT TENSION APPLICATIONS WHERE POSSIBLE. ALL FIXINGS TO BE DESIGNED USING THE FIXING MANUFACTURERS SOFTWARE. ALL FIXINGS TO BE DESIGN BY QUALIFIED PERSONNEL AND INSTALLED STRICTLY IN ACCORDANCE WITH THE MANUFACTURERS METHODOLOGY. FIXINGS TO BE TESTED ACCORDING TO STRUCTURAL SPECIFICATIONS. ALTERNATE FIXING SUPPLIERS/FIXING TYPES NOT TO BE USED WITHOUT ENGINEER APPROVAL	STRUCTURAL ENGINEER BUILDER	1	4	4	STRUCTURAL ENGINEER BUILDER
2.5	COLLAPSE OF MASONRY WALL ELEMENTS PRIOR TO COREFILLING/DURING CONSTRUCTION	3	5	15	PROVIDE ADEQUATE TEMPORARY SUPPORTS UNTIL FINAL STRUCTURE IS IN PLACE	BUILDER TEMPORARY WORKS ENGINEER	2	5	10	BUILDER TEMPORARY WORKS ENGINEER
2.6	DAMAGE TO STRUCTURE DUE TO POST-POUR PENETRATIONS	3	3	9	ALL LARGE PENETRATIONS THROUGH WALLS OR SLABS (e.g. SERVICES DUCTS) TO BE INCLUDED IN STRUCTURAL DOCUMENTATION AND INSTALLED DURING CONSTRUCTION. ALL DESTRUCTIVE WORKS TO BE APPROVED BY STRUCTURAL ENGINEER. ALL DESTRUCTIVE WORKS TO POST-TENSIONED SLABS TO HAVE APPROPRIATE CONCRETE SCANNING CONDUCTED	STRUCTURAL ENGINEER BUILDER	2	3	6	STRUCTURAL ENGINEER BUILDER
PRECAST ELEMENTS										
3.1	LIFTING AND INSTALLATION OF PRECAST CONCRETE ELEMENTS	3	5	15	WEIGHT OF PANELS TO BE MINIMISED WHERE PRACTICAL. CONTRACTOR TO ENGAGE SUITABLY QUALIFIED TEMPORARY WORKS ENGINEER TO DESIGN AND CERTIFY LIFTING METHODOLOGY	BUILDER STRUCTURAL ENGINEER TEMPORARY WORKS ENGINEER	1	5	5	BUILDER STRUCTURAL ENGINEER TEMPORARY WORKS ENGINEER
3.2	INADEQUATE PROPPING/BRACING TO PRECAST ELEMENTS	3	5	15	CONTRACTOR TO ENGAGE SUITABLY QUALIFIED TEMPORARY WORKS ENGINEER TO DESIGN AND CERTIFY PROPPING.	STRUCTURAL ENGINEER BUILDER	1	5	5	STRUCTURAL ENGINEER BUILDER
3.3	PREMATURE REMOVAL OF PROPPING TO PRECAST ELEMENTS	3	4	12	INSPECTION REQUIRED OF FINAL CONDITION AND APPROVAL FROM STRUCTURAL ENGINEER REQUIRED BEFORE DEPROPPING. ALL CLIPS TO BE INSTALLED PRIOR TO DEPROPPING.	BUILDER TEMPORARY WORKS ENGINEER	2	5	10	BUILDER TEMPORARY WORKS ENGINEER
STRUCTURAL STEEL										
4.1	COLLAPSE OR INSTABILITY OF STRUCTURAL STEELWORK DURING ERECTION	3	4	12	THE STRUCTURAL ENGINEERS DESIGN IS FOR IN SERVICE LOADS ONLY AND HAS NOT TAKEN INTO ACCOUNT THE ORDER OR METHODOLOGY OF ERECTION OF STRUCTURAL STEEL. BUILDER TO ENGAGE SUITABLY QUALIFIED TEMPORARY WORKS ENGINEER TO CERTIFY ERECTION METHODOLOGY. THIS INCLUDES EARLY SHEETING PRIOR TO THE FULL STRUCTURE BEING COMPLETE	BUILDER TEMPORARY WORKS ENGINEER	1	4	4	BUILDER TEMPORARY WORKS ENGINEER
4.2	SITE WELDING AND HOT WORKS	3	4	12	CONNECTION TYPES TO BE CONSIDERED BY THE STRUCTURAL ENGINEER IN THE DESIGN PHASE TO LIMIT THE NUMBER OF WELDED CONNECTIONS WHERE POSSIBLE. CONSIDERATION IS TO BE GIVEN TO THE LOCATION AND ACCESSIBILITY OF THE WELDING REQUIRED. THE BUILDER IS TO ENGAGE A COMPETENT BOILER MAKER AND IT IS THEIR RESPONSIBILITY TO WEAR THE ADEQUATE PPE. THE BUILDER IS TO FLAG DANGEROUS SCENARIOS WHEN THEY ARISE AND AN ALTERNATIVE CONNECTION MAY BE PROVIDED WHERE UNSAFE TO WELD.	BUILDER	1	4	4	BUILDER

DRN	AMENDMENTS	DATE	REV
NS	PRELIM STEEL ISSUE	29.05.26	A

NORTHROP
PROJECT NUMBER: 250946

Drawn: HP	Checked:
Design Coordinated: HP	Compliance Checked: --
Authorised For Issue: NS	Construction Checked: --

SKYGATE PLAY

SAFETY IN DESIGN REGISTER

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Size: A1
Revision: A

Drawing Number	Project No.	Originator	Volume	Level	Type	Discipline	Number
	25021	NOR	XX	01	DR	ST	000011

File Path: Autodesk Docs://250946 - BNE Play/25021-NOR-ZZ-ZZ-M3-ST-000001.rvt

Plot Date & Time: 29/05/2026 11:43:51 AM